NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES
KARACHI CAMPUS

Course: **Artificial Intelligence**	ASSIGNMENT- 1	**DUE DATE:** 18/02/2026
Course Code: **AI2002**		Total Questions: 6 **Total Marks: 100**

Instructions:

1. Attempt **all questions by hand**.
2. **Scan your handwritten work** and submit as a **PDF** on GCS.
3. Submit **Question 6 Source code files separately** (any language, preferably Python).
4. No plagiarism—copied work gets zero.

Question 1:

Q1. Select any one state-of-the-art Artificial Intelligence system such as the Turing Test, IBM Deep Blue, NASA's CASPER, or any modern AI system.
Answer the following in a well-structured and analytical manner:

a) Explain the core objective and theoretical foundation of the selected system. How does its design reflect the goals of Artificial Intelligence?
b) Draw a neat, well-labeled architectural diagram of the system and explain the role of each layer/module in the decision-making process.
c) Identify and discuss the key components of the system. Evaluate how effectively these components interact to produce intelligent behavior.
d) Critically assess the real-world impact of this system on research, industry, or society. In your opinion, does it represent true intelligence or narrow intelligence? Justify your answer.
e) Analyze the technical, ethical, and practical limitations of the system. How do these limitations affect its reliability and acceptance?
f) Propose future improvements or extensions that could make the system more adaptive, intelligent, and socially responsible.

Support your answers with examples, logical reasoning, and relevant references where appropriate.

Question 2:

Artificial Intelligence is widely used in different fields such as Healthcare, Education, Business, Cyber Security, Robotics, or Transportation.
Select any one field and answer the following:

a) Identify and explain a recent or emerging AI technology used in this field.
b) Describe its working mechanism and applications in detail.
c) Analyze its advantages, challenges, and ethical issues.
d) Suggest possible improvements or future developments.

Question 3

a) Write down the PEAS descriptions for the following AI Agents.
    1. AI-Based Resume Screening System.
    2. Flood Detection AI

b) Tabulate the details about the environment in which they operate. Refer to table 2.6 (AIMA book) for reference. And write a one-line reasoning for why an environment falls into the category you chose.

Question 4

Read the scenario carefully. Identify the implicit agent and formulate the problem as a state-space search problem by defining the state space, initial state, actions, transition model, goal test, path cost, and solution.

Imagine a university library where shelves are arranged in a grid, with aisles in between. Some aisles are temporarily blocked for maintenance. The library uses an automated retrieval system that starts at a fixed location and needs to fetch a specific book from a known shelf.

- The system can carry only one book at a time.
- It must reach the correct shelf, pick up the book, and stop.
- Moving around the library takes time, and unnecessary movement increases operational cost.
- The system knows the full layout of the library and the location of the requested book.

Question 5

<u>Behavior-First Tree Exploration (Uninformed Search)</u>

After an earthquake, an autonomous drone is sent into an underground cave system to search for the only exit.

The cave system consists of long tunnels, very deep paths, and several dead ends. The depth of the cave is unknown, and the location of the exit is not known in advance.

The drone has limited sensing and memory capabilities. It can only observe its current position and the immediate tunnel openings from that position. It cannot estimate how far the exit is, and it must avoid getting stuck by going too deep into dead-end tunnels.

To explore the cave, the drone follows this approach:

- It explores all possible paths up to a certain depth limit.
- If the exit is not found, it returns to the starting point.
- The depth limit is then increased slightly, and the exploration is repeated.
- This process continues until the exit is discovered.

Answer the following:

1. Which search strategy is the drone effectively using? Explain the strategy briefly in terms of how the drone explores the cave.

2. Why is this strategy appropriate in this situation? Discuss your answer with respect to:

- The drone's limited memory

- The unknown depth of the exit
- The existence of very deep dead-end tunnels

3. Draw a partial search tree (up to depth 4) to illustrate how the drone explores the cave using this strategy.

Question 6

An AI agent is placed inside an **unknown maze**. The agent can only sense its **current position** and the **immediate moves available** from that position (up, down, left, or right).

The location of the goal is **not known in advance** and may lie deep inside the maze. The maze can also contain **dead ends**, and the agent does **not** have any heuristic or distance information to guide its search.

1. Write Python code to implement:

- **Depth-First Search with a depth limit**
- **Iterative Deepening Search**

Important constraints:

- Do **not** use built-in stack or queue libraries for the main search logic.
- You may use **basic Python lists** to manage nodes.

2. Code to test your implementations on the following sample maze, represented as a 2D grid and show the output result:

```
maze = [
        ['S', 0, 0, 1, 0],
        [1, 0, 1, 0, 0],
        [0, 0, 0, 0, 'G'],
        [1, 1, 0, 1, 1]
    ]
```

Where:

- S = Start position
- G = Goal position
- 0 = Free cell
- 1 = Obstacle

3. Based on the behavior of both algorithms in this maze, **explain analytically why Iterative Deepening Search is more suitable than Depth-First Search** for this problem.